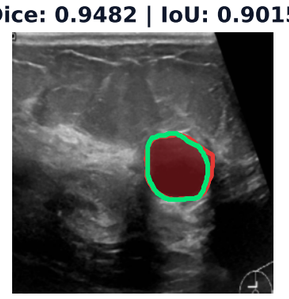
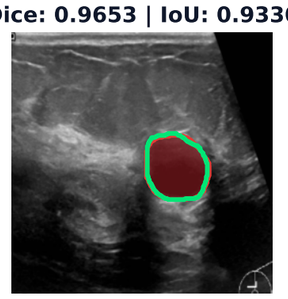
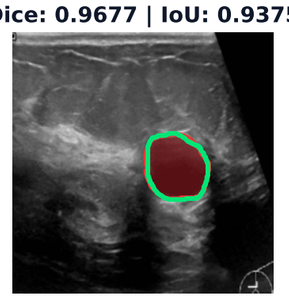
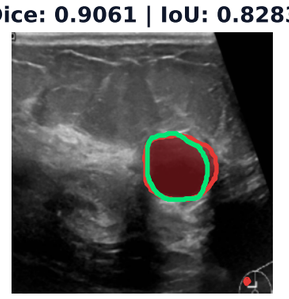
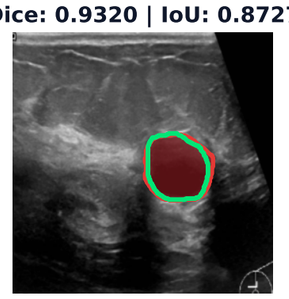
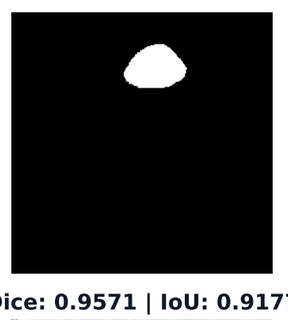
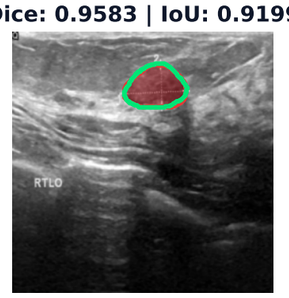
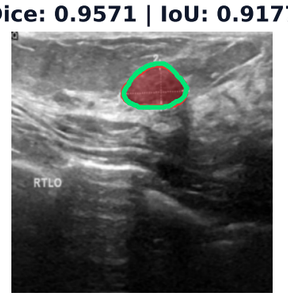
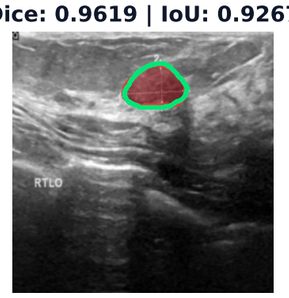
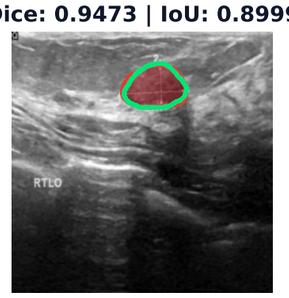
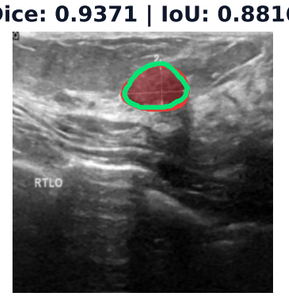
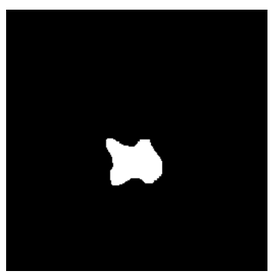
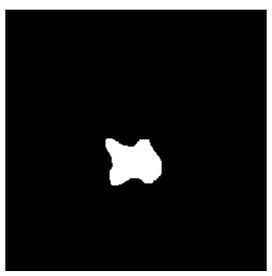
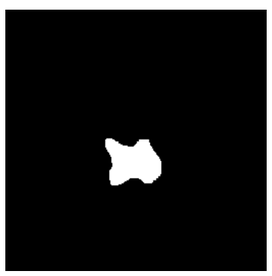
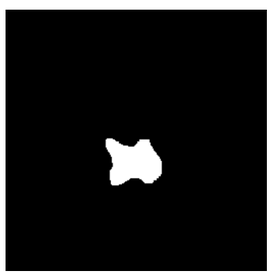
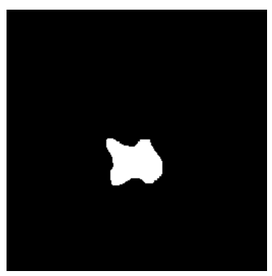
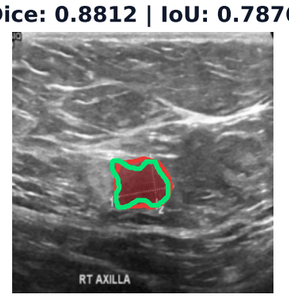
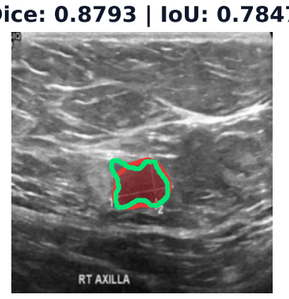
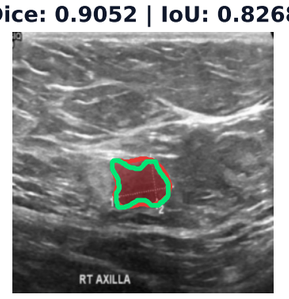
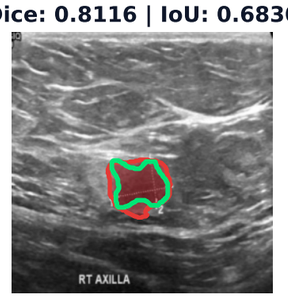
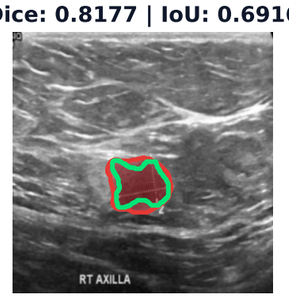
Loss-Function Qualitative Comparison: Attention U-Net Across Five Formulations (Option B)

Comparing Dice, BCE, Combined BCE+Dice (Final Model), Focal, and Tversky Loss Functions on Representative Samples

Ground Truth Boundary (GT) | Predicted Lesion Boundary | Predicted Lesion Region

1. Ground Truth (GT)	2. Dice Loss	3. BCE Loss	4. BCE + Dice Loss (Final Model)	5. Focal Loss	6. Tversky Loss
					
Original USG	Dice: 0.9482 | IoU: 0.9015	Dice: 0.9653 | IoU: 0.9330	Dice: 0.9677 | IoU: 0.9375	Dice: 0.9061 | IoU: 0.8283	Dice: 0.9320 | IoU: 0.8727
					
Benign					
					
Original USG	Dice: 0.9583 | IoU: 0.9199	Dice: 0.9571 | IoU: 0.9177	Dice: 0.9619 | IoU: 0.9267	Dice: 0.9473 | IoU: 0.8999	Dice: 0.9371 | IoU: 0.8816
					
Benign					
					
Original USG	Dice: 0.8812 | IoU: 0.7876	Dice: 0.8793 | IoU: 0.7847	Dice: 0.9052 | IoU: 0.8268	Dice: 0.8116 | IoU: 0.6830	Dice: 0.8177 | IoU: 0.6916
					
Malignant					